# Not All Local LLMs Are Equal: A Benchmark of Energy and Performance

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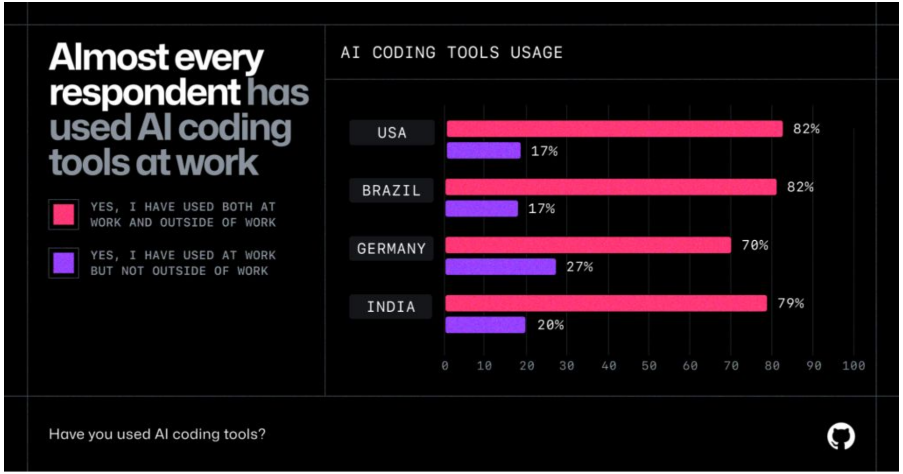
10th International Workshop on Green and Sustainable Software
April 14, 2026 - Rio de Janeiro, Brazil

# Motivation




https://github.blog/news-insights/research/survey-ai-wave-grows/#the-growing-ai-wave-in-software-development (2024)

CLIMATE AI SCIENCE

Microsoft's AI obsession is jeopardizing its climate ambitions



/ After pledging to slash its greenhouse gas emissions, Microsoft's climate pollution has grown by 30 percent as the company prioritizes AI.

by Justine Calma
May 15, 2024, 9:29 PM GMT+1

Microsoft CEO Satya Nadella delivers a speech during an event called Microsoft Build: AI Day in Kuala Lumpur, Malaysia, on May 2nd, 2024. Photo by MOHD RASFAN / AFP via Getty Images

40 Comments (All New)

https://www.theverge.com/2024/5/15/24157496/microsoft-ai-carbon-footprint-greenhouse-gas-emissions-grow-climate-pledge (2024)

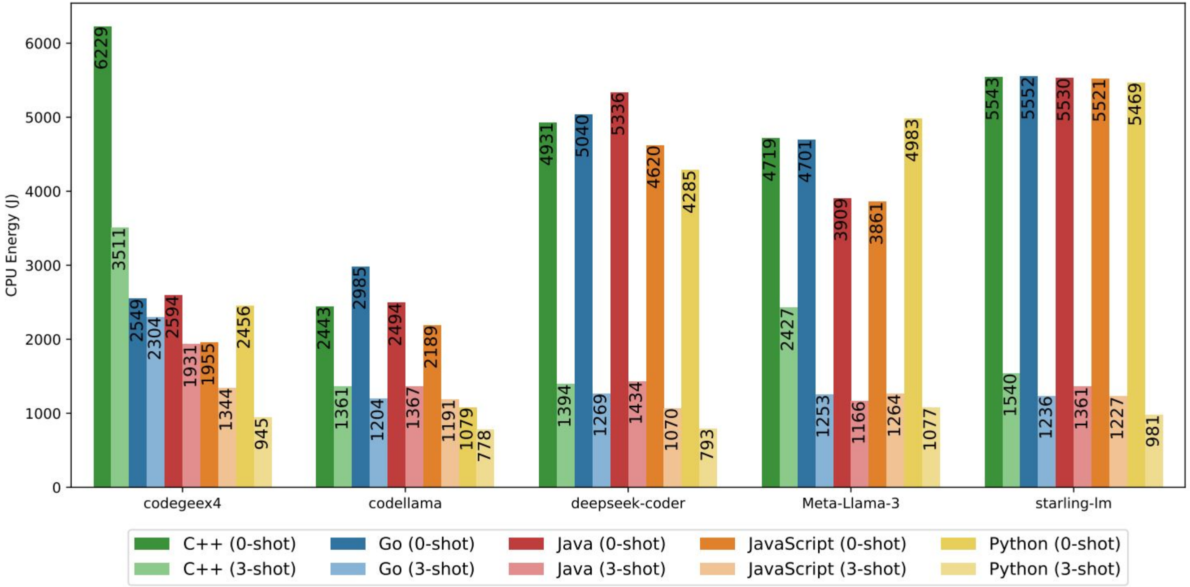
# Methodology

- 5 (Quantized) Local LLMs
  - codegeex4-all-9b
  - codellama-7b-instruct
  - deepseek-coder-6.7b-instruct
  - Meta-Llama-3-8B-Instruct
  - starling-lm-7b-alpha
- 2 LLM’s Benchmarks
  - HumanEval-X (Code Completion Tasks in C, Go, Python, Java, JavaScript)
  - MBPP+ (Code Generation Tasks in Python)
- Experimental Equipment
  - GNU/Linux OS
  - Intel Core i7-8700 CPU
  - 32GB RAM

- Cpu-only inference via llama.cpp
- Energy Measurement via CodeCarbon (uses Intel RAPL)
- OS minimal processes running

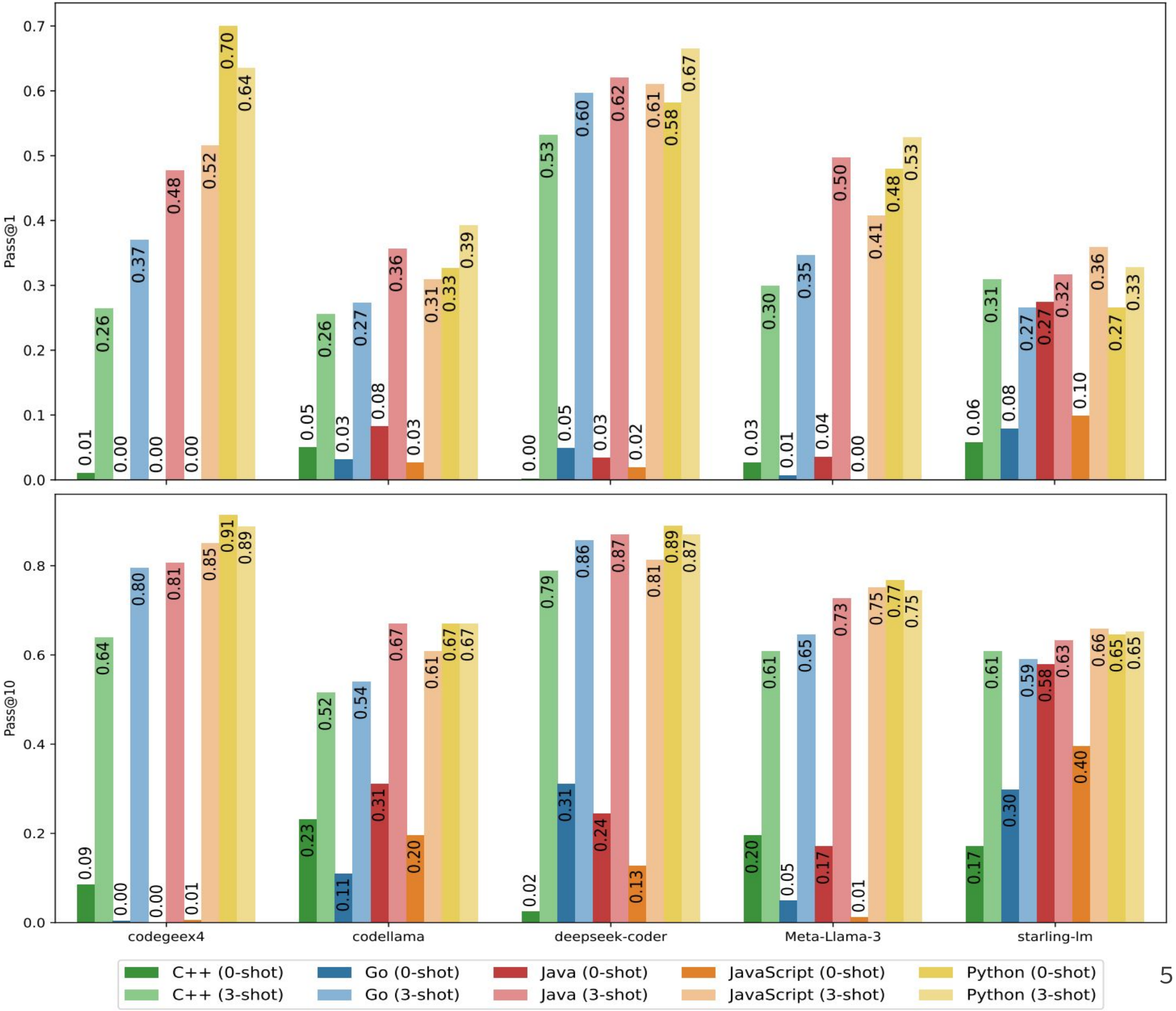
# HumanEval-X Energy Results

- Energy consumption reduction from 0-shot to 3-shot prompting



# HumanEval-X Pass@1 & Pass@10 Results

- pass@1 and pass@10 increases from 0-shot to 3-shot
- pass@10 > pass@1 in 0-shot and 3-shot



# LLMs' Energy-Based Ranking

- Ranking sorted by energy consumption
- codellama-7b-instruct is the fastest and greenest in 0-shot and 3-shot prompting
- Prompt design & model selection must be considered when deploying local LLMs

| | Rank | LLM | Energy (J) | Time (s) | Pass@1 | Pass@10 |
|---|---|---|---|---|---|---|
| 0-Shot | 1 | codellama | 1742 | 26.5 | 0.101 | 0.299 |
| | 2 | codegeex4 | $2543_{1.46\times}$ | $38.9_{1.47\times}$ | 0.138 | 0.197 |
| | 3 | deepseek-coder | $3973_{2.28\times}$ | $60.8_{2.30\times}$ | 0.143 | 0.326 |
| | 4 | Meta-Llama-3 | $4408_{2.53\times}$ | $67.3_{2.54\times}$ | 0.113 | 0.245 |
| | 5 | starling-lm | $5417_{3.11\times}$ | $82.9_{3.14\times}$ | 0.156 | 0.420 |
| 3-Shot | 1 | codellama | 1009 | 14.9 | 0.317 | 0.600 |
| | 2 | deepseek-coder | $1025_{1.02\times}$ | $15.4_{1.03\times}$ | 0.605 | 0.840 |
| | 3 | starling-lm | $1047_{1.04\times}$ | $15.7_{1.05\times}$ | 0.317 | 0.629 |
| | 4 | Meta-Llama-3 | $1186_{1.18\times}$ | $17.7_{1.19\times}$ | 0.413 | 0.694 |
| | 5 | codegeex4 | $1617_{1.60\times}$ | $24.5_{1.64\times}$ | 0.448 | 0.792 |

# Thank you for your attention!

in simaocunha71

GREENS'26

Home Architecture Results Ranking

## Not All Local LLMs Are Equal: A Benchmark of Energy and Performance

This is the online appendix for the research paper "Not All Local LLMs Are Equal: A Benchmark of Energy and Performance".

System Architecture | Complete Set of Results | LLM's Ranking


## Abstract

The rapid adoption of Large Language Models (LLMs) is transforming research, education, software development and everyday life. As their use grows, so does the diversity of available models, from general-purpose to code-oriented LLMs that can run both in data centers and on edge devices. Several benchmarks have emerged to evaluate their performance in code generation and completion tasks, yet their energy and time efficiency remain underexplored.

This paper evaluates five local LLMs on HumanEval-X and MBPP+ to analyze their accuracy, runtime and energy consumption under CPU-only inference, reflecting realistic on-device deployment scenarios where GPUs are unavailable. The results reveal clear trade-offs between effectiveness and efficiency: while some models achieve higher accuracy, others deliver comparable results with substantially lower energy use. In particular, 3-shot prompting consistently improves runtime and energy efficiency compared to 0-shot, without sacrificing code quality. These findings emphasize that prompt design and model selection must be considered together when deploying LLMs for coding tasks and call for the creation of practical prompt-efficiency guidelines to support more sustainable and efficient use of local LLMs.



**Keywords:** Large Language Models, Energy Consumption, Green Software, Programming Languages